

# Requirements Specification: Student Project Portfolio & Showcase App

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**Context:** Problem Statement #04: Student Project Portfolio & Showcase App. Centralized expo showcase platform where student teams submit multimedia artifacts and verified repository links, panel judges evaluate projects using multi-criteria digital rubrics, and authenticated public visitors participate in voting with strict anti-sybil protections and real-time leaderboard calculations.

## Requirements Table

| Req ID  | Type          | Description                                                                                                                                                                                                           | Priority | Acceptance Criteria                                                                                                                                                                                                                                                 | Rationale                                                                                                                        |
|---------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| FR-001  | Functional    | The system shall enable student teams to create and update showcase project profiles containing project title, abstract summaries, demo video links, and verified GitHub repository URLs.                             | High     | Pass: Profile is published to showcase catalog upon validation of repository link and media URL format. Fail: Invalid URL format accepted without validation.                                                                                                       | Core submission functionality enabling project artifact centralization and catalog discovery.                                    |
| FR-002  | Functional    | The system shall enable panel judges to evaluate assigned student projects against multi-criteria digital rubrics and submit finalized scores with qualitative feedback.                                              | High     | Pass: Judge can input rubric scores across all weighted evaluation criteria and successfully submit feedback, locking further edits once finalized. Fail: Submitting score without filling mandatory rubric criteria or allowing post-deadline tampering.           | Essential grading workflow enabling objective, standardized evaluation by faculty and industry panel judges.                     |
| FR-003  | Functional    | The system shall enable authenticated public visitors to cast votes for eligible showcase projects with automated anti-sybil validation checks (device fingerprinting, IP rate-limiting, and CAPTCHA).                | High     | Pass: Valid voter vote is recorded once per project category, updating voter tally. Fail: Multiple votes cast from the same verified identity/device or bot activity successfully submitting ballots.                                                               | Core engagement mechanism ensuring public choice awards remain fair, authentic, and manipulation-proof.                          |
| FR-004  | Functional    | The system shall compute and publish real-time dynamic leaderboards and category rankings based on weighted judge evaluations and verified public votes.                                                              | Medium   | Pass: Leaderboard updates category standings within 5 seconds of score finalization or voting batch aggregation. Fail: Out-of-sync standings or exposure of unfinalized judge scores.                                                                               | Provides live engagement and transparency for expo attendees, student participants, and event organizers.                        |
| FR-005  | Functional    | The system shall allow student teams to view comprehensive evaluation analytics, including rubric breakdown scores, anonymized judge remarks, and public vote counts post-expo.                                       | Medium   | Pass: Student dashboard displays authorized score summaries and feedback reports once embargo is lifted by event admins. Fail: Premature leakage of judge identity or unapproved score release.                                                                     | Facilitates constructive academic feedback, portfolio enhancement, and post-expo learning outcomes.                              |
| NFR-001 | Nonfunctional | The public voting and leaderboard calculations shall update in real-time with an API response latency under 300 ms under peak expo traffic.                                                                           | High     | Pass: Benchmarking tests confirm API latency < 300 ms at 95th percentile under simulated peak load of 5,000 concurrent voting requests. Fail: API latency exceeds 500 ms or voting service drops requests.                                                          | Guarantees responsive and stable user experience during peak expo demonstration sessions without ballot dropouts.                |
| NFR-002 | Nonfunctional | The system shall enforce anti-sybil integrity and data protection by encrypting all voter credentials and judging submissions in transit via TLS 1.3 and at rest via AES-256, alongside cryptographic ballot hashing. | High     | Pass: Automated security audit verifies 100% of API endpoints enforce TLS 1.3, database tables are encrypted, and vote manipulation detection triggers on duplicate device fingerprints. Fail: Unencrypted payload transmission or undetected sybil vote injection. | Complies with academic integrity policies and cybersecurity standards, preventing vote tampering and confidential grading leaks. |